# Regional Quantitative Problem-Solving Circle

Module 4: Graph Theory Foundations (Weeks 7–8)

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## 1 Axiomatic & Theoretical Foundation

Graph theory provides the rigorous language for modeling binary relationships and discrete structures. We decouple the visual representation of a network from its underlying topological invariants, focusing on how global properties are constrained by local boundaries.

**Definition 1** (Graph & Vertex Degree)**.** An undirected graph $G = (V, E)$ consists of a finite set of vertices $V$ and a set of edges $E$, where each edge is an unordered pair of distinct vertices $\{u, v\}$. The degree of a vertex $v$, denoted $\deg(v)$, is the number of edges incident to it.

**Definition 2** (Eulerian Circuit)**.** An Eulerian circuit in a connected graph $G$ is a closed walk that traverses every edge of $G$ exactly once, starting and ending at the same vertex.

## 2 The Core Invariant / State Function

The fundamental invariant of any finite graph links local vertex degrees to the global edge count.

**Theorem 1** (The Handshake Lemma & Degree-Sum Invariant)**.** *For any finite undirected graph $G = (V, E)$, the sum of the degrees of all vertices is exactly twice the number of edges:*

$$\sum_{v \in V} \deg(v) = 2|E|$$

**Core Invariant Framework (Topological Conservation):** Because $2|E|$ is strictly an even integer, the sum of all degrees must be even. Consequently, the number of vertices with an odd degree must be even. In dynamic traversals, parity conservation dictates that any continuous path entering a vertex must eventually exit it, linking traversal existence strictly to degree parity.

## 3 Lemma & Canonical Proof

We apply the concept of transition parity to prove the necessary and sufficient conditions for the existence of an Eulerian circuit.

**Lemma 1** (Eulerian Traversal Parity)**.** *A finite, connected, undirected graph $G = (V, E)$ admits an Eulerian circuit if and only if every vertex in $V$ has an even degree.*

*Proof.* We must prove both directions of the biconditional statement.

**Forward Direction (Necessity):** Assume that $G$ possesses an Eulerian circuit. Let this circuit begin and end at some arbitrary vertex $u \in V$. As we traverse the circuit, each time the path passes through a vertex $v$ (where $v \neq u$), it enters $v$ via one edge and leaves $v$ via another distinct edge. Thus, every passage through $v$ contributes exactly 2 to the degree of $v$ utilized by the circuit. Because the circuit traverses every edge of $G$ exactly once, all edges incident to $v$ are accounted for in pairs. Therefore, $\deg(v)$ must be an even integer. For the starting vertex $u$, the initial departure and final arrival also form a pair, plus any intermediate passages, meaning $\deg(u)$ is also strictly even. Thus, every vertex has an even degree.

**Reverse Direction (Sufficiency):** Assume that every vertex in $G$ has an even degree. Because $G$ is connected and every vertex has a degree of at least 2 (assuming $E \neq \emptyset$), $G$ must contain at least one cycle. We construct an Eulerian circuit algorithmically. Start at an arbitrary vertex $v_0$ and traverse unused edges arbitrarily to form a trail. Because every vertex has an even degree, whenever we enter a vertex $v \neq v_0$, there must be an unused edge available to exit it (the parity of unused edges at $v$ remains odd after entering, so at least 1 unused edge exists). The trail must therefore terminate at the starting vertex $v_0$, forming a closed circuit $C_1$.

If $C_1$ covers all edges in $E$, the proof is complete. If not, remove all edges of $C_1$ from $G$. The remaining graph $G'$ still has vertices of strictly even degree (we subtracted exactly 2 from the degree of every vertex along the cycle $C_1$). Because the original graph $G$ is connected, there must exist some vertex $w$ belonging to both $C_1$ and $G'$ that has incident edges remaining in $G'$. Starting from $w$, we repeat the process to find another circuit $C_2$ in $G'$, and splice $C_2$ into $C_1$ at $w$. Because the graph is finite, this process of extracting and splicing cycles must strictly terminate, eventually consuming all edges and forming a single Eulerian circuit. □

## 4 Seminar Heuristics & Socratic Framework

During the whiteboard defense phase, push students on the following diagnostic cues and common pitfalls:

- **Diagnostic Cue — "When to apply?":** Whenever a problem asks to prove that a specific network configuration is impossible, or asks for the maximum number of connections, mandate the use of the degree-sum invariant $\sum \deg(v) = 2|E|$.
- **Pitfall — The Connectedness Assumption:** Students will often prove that all even degrees imply an Eulerian circuit without stating that the graph must be connected. **Socratic Prompt:** *"What if I have two separate triangles floating in space? Every vertex has degree* 2*. Does an Eulerian circuit exist?"*
- **Pitfall — Odd Degree Nodes:** Students struggle to formalize why a path cannot start and end at different nodes in a circuit. **Socratic Prompt:** *"If you enter a room and leave it, you used two doors. What happens if a room has* 3 *doors and you want to use all of them exactly once without getting trapped?"*